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Behavioral Efficiency and Residential Electricity Consumption: A Microdata Study

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Abstract: Sustainability requires that policy makers be able to use market signals to implement energy and environmental policy and that energy consumers respond rationally to these signals. Therefore, it is essential to understand how consumers' responses to market signals are formed. We propose a new model to measure behavioral efficiency in residential electricity consumption derived from the individual householder indirect utility function. This leads to a pair of simultaneous stochastic demand frontiers for electricity consumption (kWh) and power demand (kVA). Each is a function of power demand (standing) charges and energy demand (running) charges together with the net income after demand charges, the stock of appliances and household characteristics. We estimate the model using two samples of household responses, each of which represents around one percent of the total national population available, and we also pool these samples using pseudo-panel data procedures. We demonstrate how the resulting elasticity and efficiency estimates are related to the theory of behavioral agents from recent developments in behavioral economics. These developments also use the individual indirect utility function to derive propositions based on internality and hyperbolic discounting. The econometric estimates permit the calibration of the individual welfare effects of policy initiatives using carbon tax and price incentives with behavioral agents.

Keywords: behavioral economics; energy efficiency; residential electricity; stochastic frontier analysis

JEL Classification: C21; C23; D11; D12; Q41



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1. Introduction

Sustainability requires that the world pays close attention to issues of energy efficiency because of the importance of climate change. However, there is conflict over the appropriate way to measure efficiency. The traditional approach examines the ratio of energy consumed in different sectors to the level of economic activity in the sector, while the efficiency and productivity analysis approach measures the inefficiency component of the error term in the behavioral regression function appropriate for the sector. The difference is important because many factors determine the demand for energy besides the level of sectoral economic activity, and unless these are accounted for, the potential for energy efficiency change and therefore sustainability can be overestimated. This issue becomes more serious when the sector or the macro-economy experiences massive external shocks.

In this case study, the sector that we examine is residential electrical energy consumption. The innovation that is offered is the availability of a microdata sample of residential consumers. Microdata at the level of the individual consumer are rarely available but they offer considerable richness in studying the variability of economic behavior. In 2008, 3439 residential consumers in Portugal were interviewed by the EDP Serviço Universal, the EDP SU, the Portuguese Regulated Electricity Supplier, and details were recorded about their electrical energy consumption, applicable price bands, appliance ownership and personal income. A similar exercise was repeated by the EDP SU in 2014 with another group of

3464 consumers. Both samples were carefully stratified across the whole country, and each sample represents about 1% of the national population of households. This very rich set of microdata was made available for the purposes of this paper, and it is able to give a picture at ground level of the changes in electrical energy consumption which can be used to measure electrical energy efficiency in light of the Portuguese government's drive to foster overall energy efficiency improvement, one aspect of which is the trade-off between reducing the ratio of overall energy consumption to economic activity for environmental reasons and encouraging a switch to electrical energy consumption from the direct consumption of fossil fuels. Although the data are specific to Portugal, a southern European country, there are useful general lessons to learn about consumer behavior elsewhere, particularly in regard to the responsiveness of demand to income and price changes.

However, it is important to recognize that massive external economic forces were also at work at this time. In 2008, Europe, and Portugal in particular, were entering the global financial crisis, GFC, and Portugal was among a small group of eurozone countries which were compelled to undergo considerable public finance restructuring because the country had a large ratio of public debt to the GDP. Portugal is generally recognized to have been a leader in implementing the austerity and restructuring that was encouraged by the EU. At the same time, the energy sector in Portugal experienced considerable deregulation and liberalization. As we show in this paper, the Portuguese government and regulatory policy authorities made some efforts to improve the country's greenhouse gas emissions record between the periods of the two samples, and this policy initiative has continued subsequently. All of these factors make it problematic to identify the real underlying changes in energy efficiency which were at the heart of sustainability policy in Portugal. The key macroeconomic shock for Portugal over this period has been the demand-orientated GFC. More recent shocks such as COVID-19 and the war in Ukraine have both demand and supply side aspects, and so may have an even greater long-term effect.

This paper makes use of the two microdata samples from 2008 and 2014 to investigate the stochastic energy demand function approach which interprets the inefficiency component of the regression error term as a true underlying measure of energy efficiency. In applying this model, which is now regarded as more effective in evaluating energy efficiency than the energy–economic activity ratio, to our panel of microdata, we address problems of both functional representation and parametric estimation.

In contrast to much of the literature which measures energy efficiency at the macroeconomic level, the paper uses the data to analyze the behavior of thousands of individual households. To enable this, it constructs a new model of stochastic energy demand based on the consumer's indirect utility or welfare function. This is in contrast to most of the existing literature which uses the aggregate energy consumption demand over the whole economy, the household energy consumption function or the input distance function. The advantage of a model based on the indirect utility function is that observed household behavior can be directly compared to the predictions of behavioral economic theory which uses the indirect utility function. It is possible to do this because the datasets that are used include extensive data on the electricity-using appliance stocks held by individual households in our samples. The indirect utility function model allows the analysis to include recent developments in behavioral economics which offer an explanation of why individual household usage of electricity and power may display extensive inefficiency. The limitation of this approach is the necessity to model all of the likely influences on the individual household's consumption and appliance acquisition behavior. Fortunately, our data are very comprehensive on these factors. The second contribution is to show that when households face both demand and energy charges for electricity consumption, the model yields simultaneous demand functions for electrical energy consumption and power demand, each conditional on the household's ownership of appliances as well as its demographic characteristics. Finally, the paper shows how behavioral efficiency can be estimated when the energy and power demands are treated as seemingly unrelated

regressions, and allows for the fact that the actual demand charges may be endogenous to the household.

The paper begins with a review of the literature including issues of data measurement that are relevant to our empirical study in Section 2, while Section 3 discusses the models that we will apply to the data, including both the theoretical analysis and the empirical methods. Section 4 explains the datasets in detail and Section 5 analyses the empirical results. Finally, Section 6 discusses policy conclusions from the results that address sustainability issues.

2. Literature Review

2.1. Energy in a Stochastic Demand Frontier

We adopt the definition of energy efficiency first suggested by Filippini and Hunt and co-authors in a series of papers, Filippini and Hunt [1–3], Filippini, Hunt and Zoric [4], and Orea, Lorca and Filippini [5]. In their pioneering work, Filippini and Hunt refer to their approach as the idea of ‘underlying or true energy efficiency’. This concept indicates the efficient use of energy obtained from estimating a stochastic demand frontier. They express this as an indicator of the level of inefficiency in the use of energy, using an asymmetric one-sided random variable error component u .

A typical stochastic demand frontier model specifies a behavioral frontier demand function, with a multiplicative random error term, v representing measurement, sampling and specification errors, in order to measure demand efficiency, DE . Therefore, we define efficiency as $DE = \text{stochastic frontier demand} / \text{observed demand}$. This efficiency, DE , encapsulates a ratio term that is unit invariant and can be expressed as a performance percentage. It represents the relative distance from the efficient demand frontier to an empirical observation on energy use, and it is assumed to be a non-negative variable, $DE = \exp(-u) \geq 0$, randomly distributed across the economic agents observed, so that:

$$DE \equiv \exp(-u), \quad u \geq 0 \Rightarrow 0 < DE \leq 1 \quad (1)$$

The random variable, u , therefore represents inefficiency, since

$$u = -\ln DE \cong 1 - DE \quad (2)$$

The expected value of u can be estimated by decomposing its conditional mean from the combined error term, $\varepsilon = v + u$, in an empirical equation where the other component v represents the measurement, sampling and specification error in the frontier. u is assumed to be asymmetrically distributed with a non-zero mean so that there is a high likelihood that most economic agents display low values of inefficiency and a low likelihood that some economic agents display a high level of inefficiency. Filippini and Hunt and their co-authors focus on stochastic demand frontiers estimated at the aggregate level of countries or US states. However, we wish to examine data at the level of the individual household.

Weyman-Jones et al. [6] applied a general model of this form to the structured interview response survey of residential electricity-consuming households in Portugal undertaken by the EDP in 2008. Boogen [7] carried out a study at the household level of consumption. The data are for Switzerland and consist of two telephone surveys carried out by the association of Swiss electricity companies for 2005 and 2011, consisting of 962 and 908 households, respectively. Two inputs are identified: the capital stock of household appliances and electricity consumption in kilowatt-hours, while outputs are the amount of washing, the number of meals cooked at home, the number of hours spent on entertainment, the amount of hot water services, the number of rooms and the household size. We discuss this study further in the data section.

Alberini and Filippini [8] used a similar model to estimate energy efficiency at the household level by using data from the American Housing Survey, a longitudinal study of dwellings sampled every two years from 1997 to 2009. They focus on homes in metropolitan areas with populations above 100,000. Their empirical equation is essentially identical

to that in [4], where the explanatory variables include dwelling characteristics such as the number of rooms, household size and age of the home together with binary variables representing the presence of different types of central heating, water heating and air-conditioning. The elasticity results indicate a relatively high price elasticity of the order of -0.6 , and a relatively low-income elasticity of the order of 0.02 – 0.04 . Romero-Jordan and del Rio [9] used a similar approach with the Spanish Household Budget Surveys.

In Table 1, we bring together the general characteristics and behavioral efficiency results for a number of studies that use the stochastic energy demand frontier approach. The nature of the samples differs considerably as does the focus on either residential energy consumption or specific electricity consumption. Some of the samples are of aggregate consumption while others are of individual household responses, as is the case in this paper.

Table 1. Mean efficiency of residential energy or electricity consumption in selected econometric studies.

Study	Sample	Data Type	Mean Efficiency of Residential Consumption
Filippini and Hunt [1]	29 OECD countries 1978–2006	Aggregate annual energy consumption panel	0.89–0.90
Filippini and Hunt [2]	48 USA states in 1995–2007	Aggregate annual residential energy demand panel	0.89–0.95
Filippini, Hunt, Zoric [4]	27 EU states in 1996–2009	Aggregate annual residential energy consumption panel	0.83–0.94
Orca, Llorca and Filippini [5]	48 USA states in 1995–2011	Aggregate annual residential energy demand panel	0.90–0.94
Weyman-Jones, Boucinha, Feteira-Inacio [6]	Portugal household survey 2008	Structured interview and response survey of individual households' electricity consumption	0.96–0.97
Boogen [7]	Switzerland household surveys 2005 and 2011	Two telephone surveys of individual households' electricity consumption	0.77–0.82
Alberini and Filippini [8]	USA city homes in 1997–2009	Biennial longitudinal individual housing survey of energy consumption	0.83–0.90
Romero-Jordan and del Rio [9]	Spanish Household Budget surveys 2006v12	Annual Residential electricity consumption	0.92–0.95

2.2. Data Issues in the Literature

2.2.1. Electrical Energy Demand in Portugal

The number of detailed empirical studies of the recent electrical energy demand in Portugal is relatively small, and these have made use of both aggregate data and periodic household survey data. A general finding is that the income elasticity of electrical energy consumption is low, in the region of 0.4 or less, and that consumption is heavily dependent on household characteristics.

Wiesmann et al. [10] fit cross section regressions of logged electrical energy consumption per capita to municipality level data for 2001 and data from the national consumer expenditure survey for 2005–2006. Price data were not used in the regressions, and the estimates of the income elasticity of electrical energy consumption were in the range 0.05 to 0.13 . The policy conclusion was that income-based environmental policy may be ineffective while types of dwellings and appliances are important for consumption behavior.

A pair of important papers includes those by Silva, Soares and Pinho [11,12]. Both studies use the same data based on periodic cross section household response surveys carried out by the Instituto Nacional de Estatística, INE (Statistics Portugal) with the energy regulator Entidade Reguladora dos Serviços Energéticos, ERSE. The dependent variable is

the logged electrical energy consumption per household and the explanatory variables are electricity and gas prices, household income with the addition of the number of rooms per dwelling, the age of the dwelling, family size in four categories and the heating degree days in different regions. The electricity price used was a single average bill figure comprising both energy charges and power demand charges from ERSE data. There were five of these periodic surveys by the INE and ERSE between 1989 and 2011, and these were used to construct a pseudo-panel dataset of 350 cohorts. The estimated price elasticity of electrical energy consumption was in the range of -0.54 to -0.766 while the income elasticity was in the range of 0.29 to 0.33 . The authors make the point that between the periodic surveys, Portugal experienced both severe macroeconomic shocks and environmental policy reform (green tax reform) so that periodic price and income effects may also pick up these changes in the general economic environment.

2.2.2. Using Periodic Household Surveys

The papers [11,12] make use of pseudo-panel data procedures. This idea was introduced by Deaton [13] and applied to residential electricity consumption by Bernard et al. [14]. The data consisted of periodic household surveys. The key issue here is that the households included in each survey are different and therefore the data do not comprise repeated observations on the same economic agents. This fact rules out the application of standard pooled or panel data procedures to the data by tracking the behavior of the same individual households, referred to as genuine panel data by Baltagi [15]. Instead, Deaton suggests tracking cohorts constructed from the different household groups and basing the regression data on the cohort means instead of the individual households. A cohort is defined by a set of membership rules which do not change over time. In each periodic survey, the households satisfying these rules, for example, by household size or regions, are classified into the different cohorts. The means of each variable defined over each of the cohorts then become the raw data comprising the pseudo-panel. Deaton, in [13], points out that this could introduce an errors-in-variables problem, particularly where large macroeconomic shocks are involved, but this is reduced by ensuring the cohorts reach a minimum size by limiting the number of categories used to define them. Electricity is the dominant fuel in these residential panels, and data on other fuels are not readily available. In other countries' markets, e.g., the UK and parts of northern Europe, interaction with gas prices could be important, particularly where a cheap gas supply in competition with expensive electricity is a barrier to widespread electrification.

An alternative to the pseudo-panel is sample matching and this is used by Romero-Jordan and del Rio [9], who fitted a stochastic demand frontier to household electricity consumption in Spain. Statistical matching requires the selection of households with similar socio-economic, demographic and geographical characteristics across successive cross section samples. Romero-Jordan and del Rio used six waves of the Spanish household budget survey over the period of 2006–2012 comprising 22,000 respondents in each year. The technique requires a massive number of base observations over many characteristics because the selection and matching process leads to a rapid attrition of observations. They found levels of energy efficiency above 90% and some sensitivity to the price of electricity, but concluded that elasticity was low so that reliance on pricing policy to conserve energy would require price rises that would cause energy poverty, especially amongst lower-income households. However, sample matching requires successive waves of very large panels unlike the two separated cross section samples that are available here. The pseudo-panel approach is therefore the most robust for pooling in the context of this paper.

3. Materials and Methods

3.1. Modelling the Indirect Utility Function with Rational Agents

Filippini and Hunt [3] argue that inefficiency arises if there is excessive use of energy. Our purpose is to model the efficiency and inefficiency of the use of residential electricity, but electricity is not itself a final commodity or a good that enters the utility function of the

consumer. It is an input that permits the constrained optimal consumption of the allocation of final goods and services. This idea goes back at least to the contributions of Lancaster [16]. Grösche [17] argued that residential energy consumption derives from the demand for energy services, such as the demand for thermal comfort. Households produce those services with their energy commodities (e.g., heating equipment) by using a set of fuel inputs.

We assume that the final goods and services consumed by the household are represented by the vector $\mathbf{x}' = (x_1, \dots, x_m)$ with corresponding nominal prices: $\mathbf{p}' = (p_1, \dots, p_m)$. The consumer's nominal income is symbolized as $\tilde{y}$, to differentiate it from real income. However, in order to be able to consume one or more of the final goods and services, technology dictates that the consumer must use a minimum amount of input of electrical energy in kWh, $e^{\min}$. Complementary to the required input of electrical energy, there is a required power capacity in kW or kVA, which we label $q^{\min}$.

The household's minimum required electrical energy, $e^{\min}$ is determined by a number of exogenous factors that include household characteristics and assets. Often in residential energy demand studies, these factors are related to family size and the characteristics of the building which the household occupies. However, the building itself is likely to be an indirect measure of what determines energy demand. In the case of electrical energy demand, a much more direct measure is the nature of the appliances and equipment used by the household. In fact, two different households occupying identical buildings may have relatively different electrical energy demands if they hold different ranges of appliances or there are different numbers of occupants. We use $\mathbf{z}_e$ to represent a vector of these exogenous factors including family size and the ownership of different types of appliances and equipment with different power loadings. Hence, we write frontier electrical energy input, e , as

$$e = e^{\min}(\mathbf{z}_e') \quad (3)$$

There will be a complementary household subscribed power demand in kW or kVA since power demand q is a non-decreasing function of energy usage: $q = h(e); h'(e) \geq 0$. The complementary input of power capacity, consistent with the provision of the minimum input of electrical energy to sustain a particular living standard, is also determined by exogenous factors including household characteristics and assets which we label $\mathbf{z}_q$. Therefore, frontier power demand is q .

$$q = q^{\min}(\mathbf{z}_q') \quad (4)$$

The minimum amount of the input of electrical energy, $e^{\min}$, is bought at the market price, which typically in European markets has two components: an energy charge per unit of electrical energy, e (kWh) consumed, $\tilde{p}_e$, and a monthly subscribed demand charge, $\tilde{p}_q q$, which is calibrated to the level of subscribed demand (kW or kVA) corresponding to each individual consumer. The household is able to choose from a menu of demand charges by selecting a specific subscribed demand which is restricted by circuit breakers in the property. This is the case in the data that we study here. However, in some cases, e.g., in the UK, and parts of the USA, this is set for all residential consumers at a fixed level determined by the retail supplier or utility offering the price, and is referred to simply as the standing charge.

The consumer's budget constraint is therefore:

$$\mathbf{p}'\mathbf{x} = \tilde{y} - \tilde{p}_e e - \tilde{p}_q q \quad (5)$$

In Equation (5), $\mathbf{p}'\mathbf{x}$ represents the expenditure on goods, $\tilde{y}$ is nominal income, $\tilde{p}_e$ is the nominal energy price, e is energy consumption, $\tilde{p}_e$ is the nominal demand charge, q and is the subscribed power demand. The solution to the consumer's utility maximization problem given the existence of the vectors of exogenous conditioning variables, $\mathbf{z}_e'$ and $\mathbf{z}_q'$ produces the indirect utility function:

$$U^* = v(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') = \left\{ \max_{\mathbf{x}, e, q \geq 0} U(\mathbf{x}') : \mathbf{p}'\mathbf{x} = \tilde{y} - \tilde{p}_e e - \tilde{p}_q q; e = e^{\min}(\mathbf{z}_e'); q = q^{\min}(\mathbf{z}_q') \right\} \quad (6)$$

The solution to this consumer problem is analyzed in Appendix A, where the first-order conditions are shown to be:

$$p_j = \frac{\partial U / \partial x_j}{\partial v / \partial (\tilde{y})}; j = 1, \dots, m \quad (7)$$

$$\tilde{p}_e = \frac{\partial v / \partial e}{\partial v / \partial (\tilde{y})} \quad (8)$$

$$\tilde{p}_q = \frac{\partial v / \partial q}{\partial v / \partial (\tilde{y})} \quad (9)$$

Each of the prices is set equal to the marginal benefit to the consumer of the product or the electrical energy input or power input expressed in terms of a money metric of individual welfare given by the marginal indirect utility of money income.

The demand functions derived from the interior optimum with binding constraints are

$$x_j^* = x_j^*(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e', \mathbf{z}_q') \quad j = 1 \dots m \quad (10)$$

$$e^* = e^{\min}(\mathbf{z}_e') = e^*(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e', \mathbf{z}_q') \quad (11)$$

$$q^* = q^{\min}(\mathbf{z}_q') = e^*(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e', \mathbf{z}_q') \quad (12)$$

Equation (10) shows the usual consumer demand functions for goods and services entering the utility function, while Equations (11) and (12) show the demands for electrical energy and power capacity or the subscribed demand conditional on the nominal price of all goods, energy charges, demand charges, money income and the exogenous variables that determine the technological requirement for electrical energy. The consumer's net income available to spend on final goods and services is $\tilde{y} - \tilde{p}_e e - \tilde{p}_q q$, i.e., the money income minus the expenditure on the energy input comprising energy charges and demand charges. If the consumer's actual use of energy, exceeds the efficient minimum required, if $e > e^{\min}$, the net income available for final goods and services is reduced and therefore utility, which is increasing in $\tilde{y} - \tilde{p}_e e - \tilde{p}_q q$ in the indirect utility in Function (11), is reduced as well:

$$e > e^{\min}(\mathbf{z}_e') \Rightarrow v(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e', \mathbf{z}_q') = U < U^* \quad (13)$$

Following Equations (1) and (2), define the ratio of minimum electrical energy requirement to the actual electrical energy used as behavioral efficiency, $BE_e \leq 1$, then,

$$e / e^{\min} = (1 / BE_e) \geq 1 \quad (14)$$

The inverse of behavioral efficiency, BE_e , therefore measures the extent to which the household's electricity consumption exceeds the level that would maximize the rational consumer's welfare at the same prices, income levels, family characteristics and appliance ownership.

Now, write $(1 / BE_e)$ as an exponential function of the random error component 'electrical energy inefficiency', u_e , where $u_e \geq 0$, so that:

$$e = e^{\min} \times (1 / BE_e) = e^{\min} \exp(u_e) \quad (15)$$

Now, using the demand function for energy at the interior optimum (11), we replace the expression $e^{\min}$ to obtain:

$$e = e^*(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e', \mathbf{z}_q') \exp(u_e) \quad (16)$$

In this way, we are able to interpret household behavior towards energy efficiency against the benchmark of an indirect utility frontier, where the frontier function is the demand function for the minimum energy which is required to enable the consumer to enjoy the utility from the consumption of final goods.

There is a similar analysis for the case where the consumer's subscribed demand exceeds the minimum efficiency required, so that we have the analogous expression for the behavioral efficiency of the subscribed power demand:

$$q/q^{\min} = 1/BE_q \quad (17)$$

Applying the previous argument:

$$q = q^*(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e', \mathbf{z}_q') \exp(u_q) \quad (18)$$

Taking logs and adding an idiosyncratic error component, $\exp(v_e)$, we have the familiar stochastic frontier analysis of the energy demand function with the difference that we do not need to assume that e yields direct utility to the consumer.

$$\ln e = \ln e^*(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}; \mathbf{z}_e') + v_e + u_e \quad (19)$$

Using P , the personal consumption deflator to represent the vector of the prices of final goods and services, $\mathbf{p}$, we will represent the demand function in practice as:

$$\ln e = \ln f^e(\tilde{p}_e/P, (\tilde{y} - \tilde{p}_q q)/P; \mathbf{z}_e') = \ln f^e(p_e, (y - p_q q); \mathbf{z}_e') \quad (20)$$

In (20), energy demand is a function of the energy charge in constant prices, the net income of the household after the payment of demand charges again in constant prices and the vector of exogenous variables, $\mathbf{z}_e$, that shift the household energy demand function. Then, the equation to be estimated for household electrical energy demand is

$$\ln e = \ln f^e(p_e, (y - p_q q); \mathbf{z}_e') + v_e + u_e \quad (21)$$

This is our version of the Filippini and Hunt demand function for residential energy. We note that our derivation emphasizes individual household behavior and treats the issue of multiple goods and services directly. The inefficiency component u is a random error with a positive mean $E(u_e) > 0$. Consequently, in the standard stochastic frontier analysis, it is fitted as a 'cost' function. The fitted function $\ln f^e(p_e, (y - p_q q); \mathbf{z}_e')$ will be in either double-log form or translog form.

The energy consumption in kWh given by Equation (20) will in turn determine a household subscribed power demand in kW or kVA since power demand q is a non-decreasing function of energy usage: $q = h(e); h'(e) \geq 0$. Therefore, we arrive at a complementary subscribed power demand function:

$$\ln q = \ln f^q(p_e, (y - p_q q); \mathbf{z}_q') + v_q + u_q \quad (22)$$

In Equations (21) and (22) we distinguish the component error terms $(v_e + u_e)$ and $(v_q + u_q)$, and the vectors of exogenous variables which form the full set of exogenous variables in the model: $\mathbf{z}' = (\mathbf{z}_e' : \mathbf{z}_q')$.

Our primary interest is in the income elasticity of electrical energy demand, the price elasticity of electrical energy demand, and the extent of behavioral efficiency for which we use the mean behavioral efficiency from the frontier estimation, $\overline{BE}$. However, it is critical to note that the price to the consumer has two components, the energy charge (in Euros per kWh in our dataset), p_e , and the subscribed demand or standing charge (in Euros per month in our dataset). The elasticity definitions are shown in Table 2. In the case of appliances, define the percentage effect as the $100 \times$ the proportionate or relative change in energy demand when the ownership or installation status of the appliance changes from zero to one.

Table 2. The elasticity and efficiency formulae used in calculating the empirical results.

Elasticity	Formula
1. Electrical energy income elasticity of consumption	$E_{e,y} = (\partial \ln e / \partial \ln (y - p_q)) \times (y / (y - p_q q))$
2. Electrical energy price elasticity of consumption	$E_{e,p_e} = \partial \ln e / \partial \ln p_e$
3. Electrical energy demand or standing charge elasticity of consumption	$E_{e,p_q} = \partial \ln e / \partial \ln p_q = -(\partial \ln e / \partial \ln (y - p_q q)) \times (p_q q / (y - p_q q))$
4. Subscribed power income elasticity of consumption	$E_{q,y} = (\partial \ln q / \partial \ln (y - p_q q)) \times (y / (y - p_q q))$
5. Subscribed power energy price elasticity of consumption	$E_{q,p_e} = \partial \ln q / \partial \ln p_e$
6. Subscribed power demand or standing charge elasticity of consumption	$E_{q,p_q} = \partial \ln q / \partial \ln p_q = -(\partial \ln q / \partial \ln (y - p_q q)) \times (p_q q / (y - p_q q))$
7. The percentage change effect on the energy consumption of appliance ownership	$E_{e,z} = 100 \times [\exp(\partial \ln e / \partial z) - 1]$
8. The percentage change effect on the power demand of appliance ownership	$E_{q,z} = 100 \times [\exp(\partial \ln q / \partial z) - 1]$
9. Mean behavioral efficiency of energy consumption	$\overline{BE}_e = 1 / N \sum_n E(\exp(-u_{en} v_{en} + u_{en}))$
10. Mean behavioral efficiency of power demand	$\overline{BE}_q = 1 / N \sum_n E(\exp(-u_{qn} v_{qn} + u_{qn}))$

3.2. Amending the Model for Behavioral Agents and Overconsumption

In our study, we must also address the questions of why the consumer's actual use of energy and subscribed power demand may exceed the efficient minimum. We explore the contribution of behavioral economics to this issue. Decker [18] provides a wide-ranging survey of results in behavioral economics while Farhi and Gabaix [19] offer a detailed model of the contrast between the indirect utility functions of rational and behavioral agents. Decker distinguishes between standard and non-standard consumers, while Farhi and Gabaix refer to rational and behavioral agents. The standard consumer or rational agent displays the rational calculating behavior that we illustrated in deriving the demand functions in Equations (11) and (12). However, the non-standard consumer or behavioral agent may not have stable or consistent preferences, may display various biases in the expectation of future utility, may be unwilling to actively compare different product offerings and may not be a utility-maximizer. Farhi and Gabaix identify the issues of internality and hyperbolic discounting which may be particularly important in energy demand. Internality refers to the long-term benefit or cost to the agents themselves that they fail to consider when deciding to consume a good or service. Hyperbolic discounters, according to Laibson [20], use a high discount rate over short horizons and a low discount rate over long horizons, so that there is a conflict between today's preferences and the preferences that will be held in the future. This leads to the procrastination of efficient decisions. These biases are observed in laboratory simulations but also in interviews with focus groups of energy consumers, according to Hallin et al. [21].

Applying the analysis in Farhi and Gabaix (2020), we amend our rational consumer model as follows. The behavioral agent is assumed to maximize a perceived utility function to make consumption and energy input decisions that satisfy the budget constraint in Equation (5) but that also incorporate behavioral biases so that they do not fulfill the other optimality conditions in Equations (7)–(9) that apply to the rational consumer. Instead, Farhi and Gabaix introduce the concept of a behavioral wedge. This is expressed by the difference between the price of a commodity and the marginal utility (expressed in a money metric, as captured by $\partial v / \partial(\tilde{y})$). The behavioral wedge is zero for the rational agent but nonzero for the behavioral agent. Therefore, the demand functions for each commodity and the inputs of energy and power demand satisfy:

$$\tau_j = p_j - \frac{\partial U / \partial x_j}{\partial v / \partial(\tilde{y})}; j = 1, \dots, m \quad (23)$$

$$\tau_e = \tilde{p}_e - \frac{\partial v / \partial e}{\partial v / \partial(\tilde{y})} \quad (24)$$

$$\tau_q = \tilde{p}_q - \frac{\partial v / \partial q}{\partial v / \partial(\tilde{y})} \quad (25)$$

For example, if the behavioral agent consumes excess electrical energy, this will be shown by the money metric marginal indirect utility for energy being below the market price of energy. The divergence of the money value of marginal utility from the market price reflects the failure of the agent to internalize all the costs of their choice. Therefore, the key to the measurement of τ_e and τ_q is the extent to which the agent is estimated not to internalize all the costs of their consumption decisions. In Equations (17)–(20), we have shown how this will appear as behavioral inefficiency. Therefore, we write:

$$\frac{\partial v / \partial e}{\partial v / \partial(\tilde{y})} / \tilde{p}_e = BE_e; \quad \frac{\partial v / \partial q}{\partial v / \partial(\tilde{y})} / \tilde{p}_q = BE_q \quad (26)$$

Consequently,

$$\tau_e = (1 - BE_e) \tilde{p}_e \quad (27)$$

Similarly,

$$\tau_q = (1 - BE_q) \tilde{p}_q \quad (28)$$

Gruber and Koszegi [22], also cited by Farhi and Gabaix, use a similar approach to model smokers who are hyperbolic discounters.

A key element of energy policy is the effort to restrain energy consumption and power demand by price signals, e.g., those that incorporate a carbon emissions tax in the price of energy and power demand to consumers. Farhi and Gabaix demonstrate that the welfare effects of such price changes can be measured by amending the measures of the welfare effects of price changes given by Roy's identity for the presence of the behavioral wedge. As demonstrated in Appendix A, for the rational agent, for a small price change, the money metric of the loss of welfare or real income or utility is approximately equal to the change in price multiplied by the negative of the quantity demanded:

$$\left(\frac{\partial v / \partial \tilde{p}_e}{\partial v / \partial(\tilde{y})} \right) d\tilde{p}_e = -e d\tilde{p}_e \quad (29)$$

$$\left(\frac{\partial v / \partial \tilde{p}_q}{\partial v / \partial(\tilde{y})} \right) d\tilde{p}_q = -q d\tilde{p}_q \quad (30)$$

For the behavioral agent, the welfare effects of price changes in the amended Roy's identities are:

$$\left(\frac{\partial v / \partial \tilde{p}_e}{\partial v / \partial(\tilde{y})} \right) d\tilde{p}_e = (-e - \tau_e S_{ee} - \tau_q S_{qe}) d\tilde{p}_e \quad (31)$$

$$\left(\frac{\partial v / \partial \tilde{p}_q}{\partial v / \partial(\tilde{y})} \right) d\tilde{p}_q = (-q - \tau_q S_{qq} - \tau_e S_{eq}) d\tilde{p}_q \quad (32)$$

In Equations (31) and (32), the terms S_{ee} and S_{qq} are the Slutsky substitution effects of the price changes on the consumption of energy and power, for example, $S_{eq} = \partial e / \partial \tilde{p}_q$ when the income effect of the price change has been compensated. In general, we expect own price and complementary price Slutsky effects to be negative so that the net effect is to reduce the welfare loss to the behavioral agent of a rise in price. A full derivation of these results is described in the work of Farhi and Gabaix [19], but an intuitive explanation can be given by thinking of a consumer's two-dimensional convex indifference curve diagram for true or experienced utility, as illustrated in Figure 1. The rational agent is located at A at the tangency of an indifference curve with the budget constraint line, mm , thereby fulfilling all the first-order conditions for a constrained utility maximum. The behavioral agent is located at an intersection of the budget line and a convex experienced utility indifference

curve rather than at a tangency, fulfilling only the budget constraint condition, so that they display the overconsumption of one of the commodities. For example, the behavioral agent may be located at B on the budget constraint mm , where there is an overconsumption of commodity X_1 . If the price of this commodity rises so that the budget constraint pivots inwards to mn , the behavioral agent will likely substitute away from the commodity by reducing overconsumption, locating at a point C further along the new budget line, closer to the rational agent optimum. But this represents a higher level of experienced utility shown by the dashed indifference curve through C, since the previous allocation was at a more extreme intersection of the budget line and the convex indifference curve. Clearly, the extent to which the behavioral agent can move to a higher experienced utility indifference curve as a result of the price rise reflects the substitution effect in the demand function and, hence, this enters the amended Roy's identity for the welfare effects of a price change for the behavioral agent. Thus, a price rise for energy or power demand will include a positive offset on the welfare of behavioral agents because they are incentivized to be located closer to the rational agent's optimum, i.e., to bring their consumption closer to the efficient frontier. The results shown later in this paper will be related to these comparative static effects shown in Figure 1.

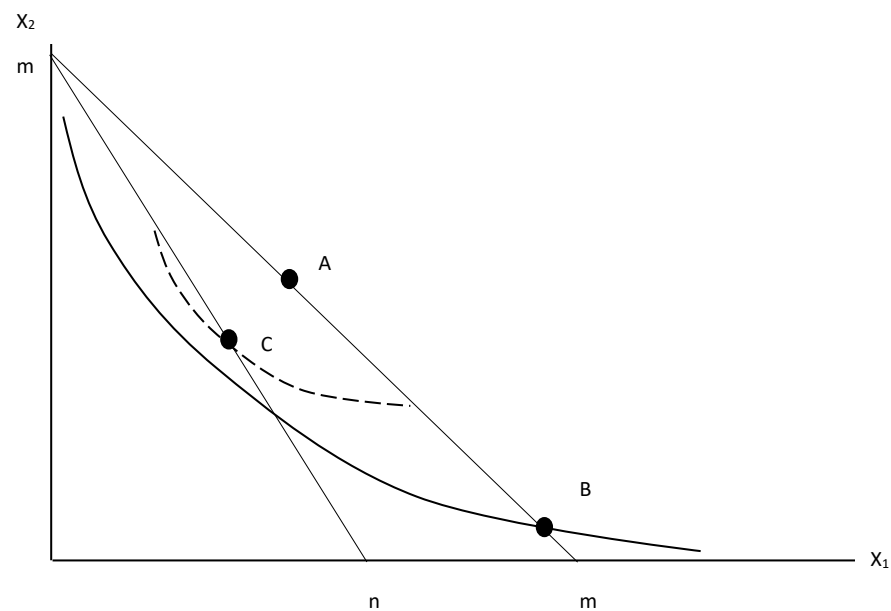


Figure 1. Illustration of rational and behavioral agent outcomes.

To extract the welfare effects of a price change, we use the well-known Slutsky equations in an elasticity format:

$$\left(\frac{\partial v / \partial \tilde{p}_e}{\partial v / \partial (\tilde{y})} \right) d\tilde{p}_e = (-e - ((1 - BE_e)\tilde{p}_e)(e/\tilde{p}_e)(E_{e\tilde{p}_e} + (\tilde{p}_e e/y)E_{ey}) - ((1 - BE_q)\tilde{p}_q)(q/\tilde{p}_e)(E_{q\tilde{p}_e} + (\tilde{p}_e q/y)E_{qy})) d\tilde{p}_e \quad (33)$$

$$\left(\frac{\partial v / \partial \tilde{p}_q}{\partial v / \partial (\tilde{y})} \right) d\tilde{p}_q = (-q - ((1 - BE_q)\tilde{p}_q)(q/\tilde{p}_q)(E_{q\tilde{p}_q} + (\tilde{p}_q q/y)E_{qy}) - ((1 - BE_e)\tilde{p}_e)(e/\tilde{p}_q)(E_{e\tilde{p}_q} + (\tilde{p}_q e/y)E_{ey})) d\tilde{p}_q \quad (34)$$

This theoretical analysis suggests that Pigouvian taxes might be used to correct for overconsumption and behavioral inefficiency. However, Farhi and Gabaix recognize two objections to this policy. First, in the model, agents make mistakes that the government can identify, but correcting for these to move consumers closer to the rational agent optimum is a form of paternalism. Additionally, governments may not understand agents' motives and constraints well and may find it politically difficult to implement taxes to improve behavioral decisions. Second, if poorer consumers are the ones making behavioral mistakes

such as internalities or hyperbolic discounting, then these taxes may worsen the income distribution, which they describe as a trade-off between internality correction and redistribution. As well as instigating regulated price and tax changes, policy makers have the option of direct controls and what have become known as nudge policies, i.e., those that guide consumer decisions through better information and advice, and provide guidance on making efficient decisions. Farhi and Gabaix [19] argue that these are particularly relevant where external effects are present, for example, in the case of a policy to restrict carbon emissions. In the empirical section of this paper, we attempt to estimate these effects statistically.

3.3. Estimation of Behavioral Efficiency

The aim is to estimate Equations (21) and (22) but there are several options available. Our estimation strategy objective is to obtain consistent and efficient estimators of the parameters including the regression elasticities and the mean and variance parameters of the composed error distributions. Each of the proposed estimation strategies does this under different assumptions about the covariance of the error terms.

We begin by treating the two equations for energy demand and subscribed power demand as unrelated and using standard stochastic frontier analysis maximum likelihood estimation (MLE-SFA) applied to each equation separately. This is applied to two cross section samples for 2008 and 2014. It can also be applied to a pooled cohort pseudo-panel after testing for the poolability of the two cross section samples.

The general stochastic frontier analysis model for a cross section sample of n observations is the regression:

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i; i = 1, \dots, n \quad (35)$$

There are two components to the error term $\varepsilon_i = v_i + u_i$: v_i is the usual regression two-sided symmetric zero-mean random variable reflecting sampling, specification and measurement error, while u_i is the error component reflecting inefficiency, and is a one-sided asymmetrically distributed random variable with a positive mean in the case of the models used here. The inefficiency component is based on the idea that inefficiency can be treated as a nonnegative random variable with most observations clustered close to the positive side of the zero value and a positive tail of observations reflecting greater values of inefficiency with diminishing probability. The original model is due to Aigner Lovell and Schmidt [23] and Meeusen and van den Broeck [24] and it has been extensively developed since then; see, for example, the studies conducted by Kumbhakar and Lovell [25] and Kumbhakar et al. [26]. The basic model makes the following distributional assumptions: $v_i \sim i.i.d.N(0, \sigma_v^2)$ and $u_i \sim i.i.d.N^+(0, \sigma_u^2)$ so that the log-likelihood function of the sampled y_i with respect to the parameters $\boldsymbol{\beta}, \sigma, \lambda$ can be constructed as follows, where $\sigma^2 = \sigma_u^2 + \sigma_v^2$, and $\lambda = \sigma_u / \sigma_v$ and $\Phi(-\varepsilon_i \lambda / \sigma)$ is the cumulative distribution function of the standard normal distribution from the composed normal and half-normal error terms.

$$\ln L(y|\boldsymbol{\beta}, \sigma, \lambda) = \text{constant} - n \ln \sigma + \sum_{i=1}^n \ln \Phi(-\varepsilon_i \lambda / \sigma) - (1/2\sigma^2) \sum_{i=1}^n \varepsilon_i^2 \quad (36)$$

Maximizing this log likelihood function by numerical methods after substituting for the ε_i from Equation (35) allows us to estimate the parameters and to test for the presence of inefficiency. If λ does not differ significantly from zero, then this implies $\sigma_u = 0$ and the inefficiency distribution collapses on the zero value, so we conclude inefficiency is absent. If λ does differ significantly from zero, then this implies $\sigma_u > 0$, and inefficiency is present in the sample. We have already defined behavioral efficiency for energy consumption in Equation (15) as $e = e^{min} \times (1/BE_e) = e^{min} \exp(u_e)$. A measure of inefficiency can then be based on the conditional expected value from estimation: $E(u_i|\varepsilon_i)$; see the studies conducted by Jondrow et al. [27] and Battese and Coelli [28]. This is one of the core models that we estimated.

However, other econometric assumptions must also be allowed for. One alternative is to treat Equations (21) and (22) as a related pair of behavioral functions so that we adopt the seemingly unrelated regression model estimated by generalized least squares, GLS-SUR. This permits the composed error terms in the two equations to be interdependent, indicating that similar random shocks are applied to both energy consumption and to power demand. Then, the feasible iterative GLS-SUR procedure produces statistically consistent estimators for the parameters β, σ . We are able then to test cross-equation restrictions on the elasticities.

The standard MLE-SFA is no longer suitable, so that now we use the quasi-maximum likelihood estimation suggested by Fan et al. [29]; see also the work of Henningsen and Kumbhakar [30] and Kuosmanen et al. [31] to decompose the residuals from the consistent GLS-SUR estimates. Since the GLS-SUR estimators have the consistency property, we use ε_i^{CON} as symbolic notation for the residuals from the GLS-SUR.

Fan et al. show that the quasi-likelihood can be written as a function of the single parameter $\lambda = \sigma_u / \sigma_v$, the signal-to-noise ratio, after replacing the error terms ε_i in Equation (36) with these residuals from the consistent estimator ε_i^{CON} as follows.

$$\ln L(\lambda) = \text{constant} - n \ln \hat{\sigma} + \sum_{i=1}^n \ln \Phi \left[\frac{-\hat{\varepsilon}_i \lambda}{\hat{\sigma}} \right] - \frac{1}{2\hat{\sigma}^2} \sum_{i=1}^n \hat{\varepsilon}_i^2 \quad (37)$$

Here,

$$\hat{\varepsilon}_i = \varepsilon_i^{CON} - \left(\sqrt{\frac{2}{\pi}} \right) \left(\frac{\hat{\sigma} \lambda}{\sqrt{(1 + \lambda^2)}} \right) \quad (38)$$

And,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\varepsilon_i^{CON})^2 / \left(1 - \left(\frac{2}{\pi} \right) \left(\frac{\lambda^2}{1 + \lambda^2} \right) \right) \quad (39)$$

Use (38) and (39) to substitute out $\hat{\varepsilon}_i$ and $\hat{\sigma}$ from (37). Maximize (37) by a grid search over values of the single parameter λ to find the quasi-likelihood estimate $\hat{\lambda}$ that maximizes (37), and insert estimate $\hat{\lambda}$ into (38) and (39) to obtain (new) estimates of $\hat{\varepsilon}_i$ and $\hat{\sigma}$. Subsequently, calculate $\hat{\sigma}_u$ and $\hat{\sigma}_v$. Therefore, by applying the feasible GLS-SUR estimator to the pair of energy demand and power demand functions, and subsequently operating the Fan et al. [29] procedure, we can obtain a new set of behavioral efficiency scores.

A third approach recognizes that Equations (21) and (22) might form a simultaneous equation system in which the endogeneity of one or more explanatory variables arises, so that the explanatory variables are correlated with the error term. One variable in particular may be a problem, and that is the consumer's income after demand charges since this might be determined by the consumer's reaction to the published demand charges. To address this issue, we use instrumental variable estimation with two-stage least squares, IV-2SLS. Again, to measure behavioral efficiency, we use the quasi-maximum likelihood estimation suggested by Fan et al. [29] to decompose the residuals from the consistent IV-2SLS model. Therefore, we apply Equations (38) and (39) again where, this time, the residuals from the consistent estimators ε_i^{CON} are those from applying the IV-2SLS estimator. An alternative procedure is the Karakaplan and Kutlu [32] instrumental variables treatment of the MLE-SFA estimator, used by Romero-Jordan and del Rio [9].

Several types of parameter estimates are produced for $E_{e,y}$, E_{e,p_e} , E_{e,p_q} , $E_{q,y}$, E_{q,p_e} , E_{q,p_q} , $\overline{BE_e}$, $\overline{BE_q}$ depending on the sample: the cross section for 2008, (XS08), the cross section for 2014, (XS14), or the pooled pseudo-panel (PP) from each of three estimation strategies: MLE-SFA, GLS-SUR, and IV-2SLS. We have computed the results from all three modeling approaches, and it turns out that the results are very similar across all three approaches so that in the text we report only one set of results in full but comment on all three sets of results when we report the elasticities and the behavioral efficiency scores.

4. Data

The data are discussed in the next three subsections covering household and appliance ownership characteristics, the price of electricity and the censoring of the consumption data. As was explained in Section 1, in 2008, 3439 residential consumers in Portugal were interviewed by EDP SU, recording their electrical energy consumption, applicable price bands, appliance ownership and personal income. The exercise was repeated by EDP SU in 2014 with another group of 3464 consumers. Both samples were carefully stratified across the whole country, and each sample represents about 1% of the national population of households. We first eliminated responses with incomplete data in any category. The largest number of the incomplete responses concerned the non-disclosure of net family income and a few related to the non-disclosure of appliance ownership. The final number of usable and complete household responses across all variables was 2622 in 2008 and 2519 in 2014. To implement the pseudo-panel procedure, we first identify key characteristics of the sample observations in each cross section. The characteristics chosen for our samples were the family size, region, and subscribed demand class. Then, we calculate the mean values of all variables for each group that shares these characteristics. These mean values make up the pseudo observations which can be pooled across the two separate years for each group.

4.1. Household Characteristics and Appliance Ownership

The household characteristics recorded in the surveys are shown in Table 3 and the appliance ownership data recorded are shown in Table 4, where appliances or equipment with a high kVA power rating are designated as an HP rating. Technological advances in residential electricity use which may improve efficiency include smart metering, remote wireless household control of appliances and access to virtual energy networks, see Decker [18].

Table 3. Household characteristics recorded in the responses to the EDP SU surveys.

Household data	Unit
Region	Name of region
Subscribed demand band	Classes 1–6
Subscribed demand level	kVA
Consumption per year	kWh
Family size	Number
Net Income per month	Euros per month in 2014 prices

Source: EDP SU residential consumer data surveys conducted in 2008 and 2014.

Table 4. Appliance ownership data by type of variable and rating of High Power (HP rating) recorded in the EDP SU surveys.

Appliance Data	Type of Variable with High Power Appliances Identified by HP Rating	
Fan N°	Number	
Fan ownership	Binary (1,0)	
Central heating ownership	Binary (1,0)	HP rating
Local heating N°	Number	
Local heating ownership	Binary (1,0)	
Water heater ownership	Binary (1,0)	HP rating
Air conditioning N°	Number	
Air conditioning ownership	Binary (1,0)	HP rating

Table 4. Cont.

Appliance Data	Type of Variable with High Power Appliances Identified by HP Rating	
Microwave ownership	Binary (1,0)	
Fridge N°	Number	
Fridge ownership	Binary (1,0)	
Freezer N°	Number	
Freezer ownership	Binary (1,0)	
Washing machine N°	Number	
Washing machine ownership	Binary (1,0)	
Clothes' dryer ownership	Binary (1,0)	HP rating
Washing + drying ownership	Binary (1,0)	HP rating
Dish washer N°	Number	
Dish washer ownership	Binary (1,0)	
TV N°	Number	
TV ownership	Binary (1,0)	
PC N°	Number	
PC ownership	Binary (1,0)	
Printers N°	Number	
Printers ownership	Binary (1,0)	
Vacuum cleaner N°	Number	
Vacuum cleaner ownership	Binary (1,0)	
Iron N°	Number	
Iron ownership	Binary (1,0)	
Stove N°	Number	
Stove ownership	Binary (1,0)	
Oven N°	Number	
Oven ownership	Binary (1,0)	

Source: EDP SU residential consumer data surveys conducted in 2008 and 2014.

The surveys are analyzed in much greater detail and the trends are discussed in Boucinha et al. [33]. The data on the market penetration of the appliance types are shown in Table 5 and graphically illustrated in Figure 2. We show the market penetration scores from two sources: the samples used for the data and the population average estimated by EDP SU. We see, for example, that the market penetration of central heating is higher amongst the sample responses than in the population at large; the same is true for several other appliance types, suggesting the sample data reflects consumers who are slightly more electricity-intensive users than the population as a whole.

Ownership is generally stable: central heating, washing machines, televisions and stoves are each almost identical in market penetration between 2008 and 2014. On the other hand, some appliance types exhibit considerable market penetration changes. Fans, local heating, clothes drying, in particular washing and drying, and printer ownership show considerable declines over the period of 2008 to 2014. In the same period, water heating ownership has expanded considerably. Effects on kWh consumption, however, will vary since the maximum power ratings (kW or kVA) of these appliances vary substantially. Central heating (electric), clothes drying and air conditioning consume high levels of power, in the region of 4 kW. Fridge/freezers which may operate for many hours per day generally have low power ratings in the range of 0.25 to 0.5 kW, while water heating is

usually rated about 2 kW. Of course, appliances are not normally running continuously or simultaneously and Table 6 reports the average load factors for households in 2008 and 2014. This table also shows that while the mean subscribed demand increased a little between 2008 and 2014 by less than 1 percent, the consumption and load factor dropped by almost 25 percent in each case—a factor consistent with the massive macroeconomic GDP shock experienced by the Portuguese economy and consumers in particular between 2008 and 2014.

Table 5. Market penetration of appliances (percents) recorded in EDP SU surveys and in national estimates.

Data Comparison	Sample		Population Estimates	
Appliance Type	2008	2014	2008	2014
Fan	43.2	29.3	43.9	30.4
Central heating	6.9	6.9	2.6	2.7
Local heating	44.4	29.1	47.8	31.6
Water heater	19.4	23.9	13.8	14.2
Air conditioning	30.5	29.6	12.5	13.8
Microwave	89.8	89.2	88.1	84.9
Fridge	99.8	99.8	99.4	99.6
Freezer	62.4	59.2	55.5	47.0
Washing machine	94.7	94.7	94.6	91.2
Clothes dryer	36.4	29.7	27.4	16.1
Washing + drying	5.8	2.9	5.6	2.4
Dish washer	73.0	67.3	58.0	44.4
TV	98.4	99.4	99.6	98.8
PC	76.0	72.9	71.7	62.2
Printers	67.0	53.5	62.2	37.2
Vacuum cleaner	94.0	87.6	92.5	83.1
Iron	98.4	95.0	98.0	92.8
Stove	37.5	37.7	27.3	20.5
Oven	65.4	68.7	57.2	52.9

Source: Authors' calculations from EDP SU residential consumer data surveys conducted in 2008 and 2014.

Table 6. Load factors in the household consumption data from the EDP SU surveys using industry average power factors.

Average Load Factor Calculation	2008	2014	Percentage Change
Mean annual consumption, kWh	7932.64	5952.17	−24.97
Mean subscribed demand, kVA	14.41	14.51	0.72
Assumed average power factor	0.6	0.6	
Mean household load factor	0.10	0.08	−25.50

Source: Authors' calculations from EDP SU residential consumer data surveys conducted in 2008 and 2014. The power factor in row 4 of Table 6 converts kVA to kW using a standard industry average power factor for domestic appliances of 60 percent, see Lund Instrument Engineering Inc., Orem, Utah, USA <https://www.powerstream.com/VA-Watts.htm>, accessed 2 April 2024.

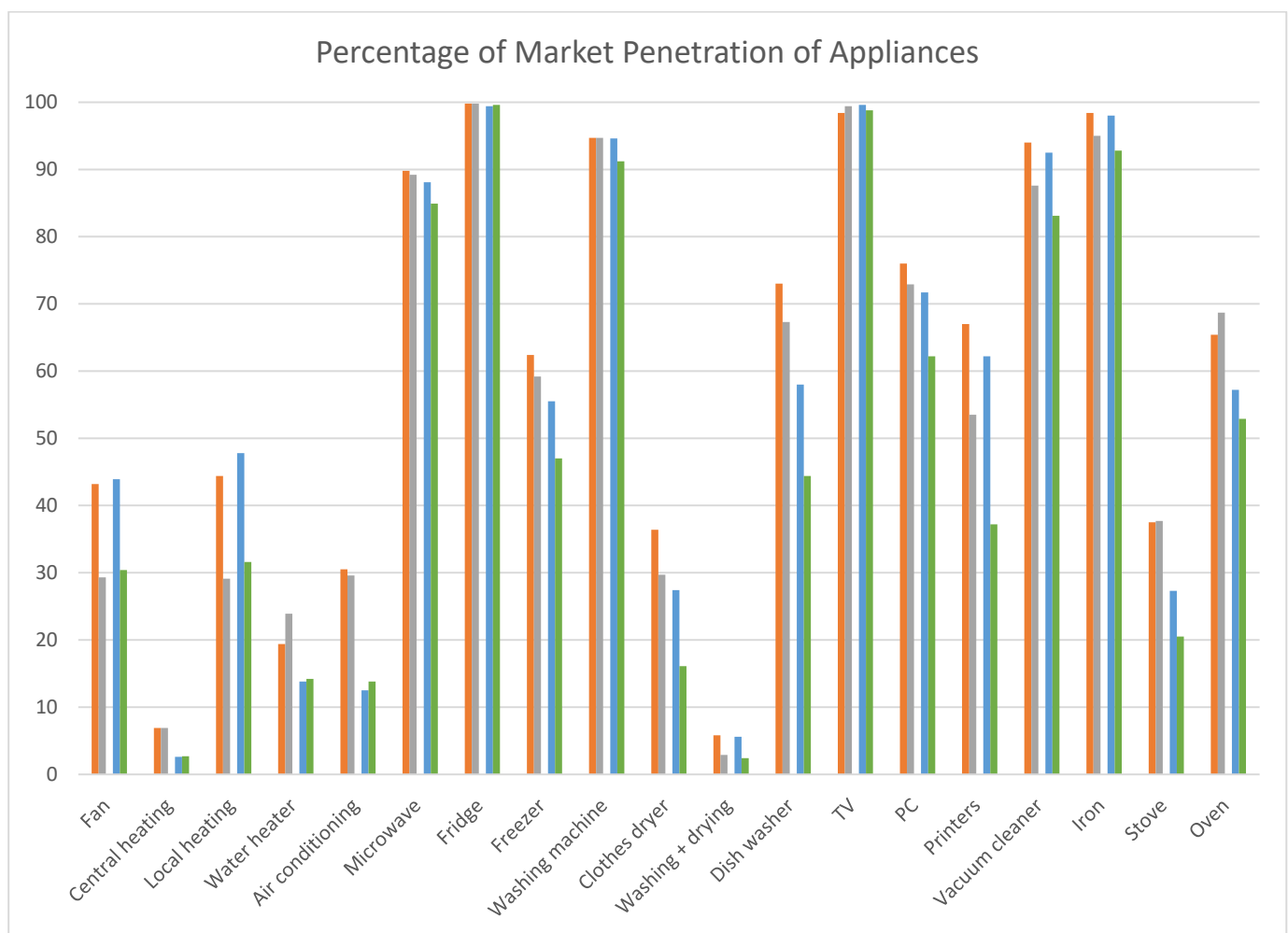


Figure 2. Percentage of market penetration of appliances. Key: for each appliance type, two leftmost bars, (brown, grey) are EDP SU survey estimates for 2008 and 2014, respectively; two rightmost bars, (blue, green) are population estimates for 2008 and 2014, respectively.

The implications for modeling consumer demand in the individual cross section samples and the pseudo-panel data are therefore complex, and the major changes in consumer behavior between 2008 and 2014 make it possible that there could be differences between the observed behavior in each of the cross section samples and the cohort mean panel. A key feature, however, is the likelihood of an important impact from the ownership of high-powered appliances.

4.2. The Price of Electricity for Residential Consumers

As we discussed above, it is important to be able to construct an appropriate cost of electricity for residential consumers. Tariff schedules available to residential consumers in Portugal are relatively complex compared with those in many other countries, e.g., those in the UK. In the 2008 sample, virtually all residential consumers faced the same choice of tariff schedules from the regulated supplier, EDP Serviço Universal, but by 2014, several residential consumers had changed their supplier, and EDP Serviço Universal served about 40 percent of consumers. The structure of the tariffs, however, remains relatively similar for the incumbent and the new suppliers, so that we have taken the regulated tariff schedules to be representative for both 2008 and 2014. In both years, there were three forms of tariff schedules available, for very low power load consumers (<2.3 kVA), intermediate power load consumers (≥ 2.3 kVA and <20.7 kVA) and very high power load consumers (>20.7 kVA). The numbers in the very low and very high categories are relatively low,

and therefore we concentrate on the intermediate load residential tariff as representative. Consequently, we take as the representative tariff schedule for our samples the ‘simple’ tariff structure offered to intermediate load consumers. This is still relatively complex compared to many other electricity market examples.

This intermediate load tariff structure consists of eight separate kVA subscribed demand charges in Euros per month with two corresponding energy charges in Euros per kWh. Consequently, the tariff contains both a direct income effect and a variable price effect, so that both components should be included in the analysis. In our sampling, to avoid having categories with very few consumers, we amalgamated the eight bands into six, as indicated in Table 7, which shows the demand charges in the current year’s price in Euros per month.

Table 7. Subscribed power demand charges in current year prices from EDP SU tariff schedules.

Subscribed Demand (Fixed Charge)	Load	(EUR/Month)	(EUR/Month)
Subscribed Demand Band	(kVA)	2008	2014
1	3.45	5.74	4.64
1	4.6	7.45	6.03
2	5.75	9.15	7.42
2	6.9	10.85	8.81
3	10.35	15.70	12.96
4	13.8	20.61	17.12
5	17.25	25.42	21.28
6	20.7	30.42	25.44

Together with these demand charges, there are variable energy charges as shown in Table 8, again in the current year’s prices.

Table 8. Energy consumption charges in current year prices from EDP SU tariff schedules.

Energy	2008 (EUR/kWh)	2014 (EUR/kWh)
Simple Tariff $\leq$ 6.9 kVA	0.1143	0.1528
Simple Tariff $>$ 6.9 kVA	0.1143	0.1543

In both 2008 and 2014, peak, off-peak and night energy charges were also available under the regulated tariff, but few consumers chose this option, and such options were not generally available from non-regulated suppliers. In addition, the mean cohort load factor was only eight percent, suggesting that very little load was shifted into the off-peak and night periods. The mean cohort load factor is calculated as the consumption (kWh) divided by the subscribed demand (kVA) $\times$ average appliance power factor $\times$ 8760 where the power factor converts kVA to kW and a standard industry average power factor for domestic appliances is 60 percent, see above. Consequently, we used the energy charges in the simple tariff as the estimate of the marginal kWh price for all households. It remains the case that the cost of electricity to residential consumers is not the same across all consumers in any one year and it has two components: a power demand charge which generates an income effect and a unit consumption charge for energy which will contain both an income effect and a substitution effect. We use both components as explanatory variables in the estimated regressions. In 2008, there was a single energy charge over all households, so that in the 2008 cross section it is included in the intercept term. In the case where the cross sections are combined across years, the current year’s prices are converted to 2014’s constant prices. Since we know the subscribed power demand and the energy consumption for each consumer household, we are able to compute the exact cost of electricity for each

household under the tariff schedule shown in Tables 7 and 8 with eight subscribed demand levels and up to two energy charges.

In fact, the structure of the demand charges in relation to subscribed demand is, apart from rounding effects, a linear relationship. Consequently, knowing the charges for at least two consumer classes, we can interpolate the intercepts and slopes of the demand charges directly from the tariff as the linear equations:

Demand charges for 2008 in 2014's prices:

$$\text{demand charge, Eur/month} = 0.66 + 1.51 \times (\text{subscribed demand, kVA})$$

Demand charges for 2014:

$$\text{demand charge, Eur/month} = 2.19 + 0.80 \times (\text{subscribed demand, kVA})$$

As we argue subsequently, the household is able to choose from the menu of demand charges by selecting a specific subscribed demand which is restricted by circuit breakers in the property; if the subscribed demand ceiling is too low, the household can pay to increase it. This makes the demand charge into a deterministic linear function of the household's subscribed demand, and this can be used to help to identify the behavioral parameters in an estimated econometric equation.

4.3. Censoring the Consumption Data

The household electricity consumption data are critical to this study, particularly to the measurement of efficiency. However, there are reasons for censoring the raw data at the lower and upper tails of the distribution. At the lower end of the consumption distribution, the data include unlikely, low observed values. We hypothesize that these extremely low consumption households form a different population from the majority of consumers and need to be excluded from the regression to avoid biasing the efficiency scores.

Boogen [7] identified a similar issue for her samples of Swiss consumers taken in 2005 and 2011. In this case, the surveys asked about electricity consumption for cooking, heating, lighting, entertainment, clothes washing and showers in the week preceding the survey. Minimum energy consumption levels were identified for the aggregate levels of these services and converted to an equivalent annual electricity consumption load in kWh and then observations below these minimal annual equivalent consumption levels were excluded. Boogen reported a range for the minimal values as follows, which is described as a conservative estimate for identifying the level below which outliers are excluded: for single households with no electric clothes washing or shower, the minimal value was 450 kWh per year. Adding clothes washing raised this to 500 kWh and an electric shower raised it to 850 kWh. In addition, an added family member was assumed to contribute 100–500 kWh per year depending on the level of service used. Households with lower annual consumptions were excluded as outliers.

In this paper, the different nature of our dataset means that we must address this issue with a different methodology. However, the results we arrived at to set minimal consumption levels for excluding outliers turned out to be similar in terms of the annual equivalent kWh consumption to those used by Boogen. These observations arise from the distinction between full-occupancy and low-occupancy households such as second homes or holiday homes and from the case of households which are mostly but not entirely off-grid. The 2014 survey included a question on whether the house was a first or second home, but the 2008 survey did not include this question. However, even the 2014 home occupancy response is problematic for identifying low occupancy consumption. This is because 13 percent of the designated second homes displayed consumption levels in excess of the median consumption of the designated first homes. Also, nine percent of designated second homes displayed consumption levels that were more than twice the median consumption of designated first homes. This suggests that the first/second home response has not entirely reflected the real distinction between full and low occupancy. It is

also possible that respondents are misstating or misinterpreting the second homes question. Reasons for abnormally low consumption can include low-occupancy, mainly off-grid lifestyle choices, reliance on back-up self-generation, reliance mainly on non-electrical energy for cooking, heating and water heating. In addition, we need to discover a heuristic rule that can be applied to identify the distinction in the 2008 sample.

Table 9 indicates the range of median values of electrical energy consumption by first and second homes in 2014.

Table 9. Electrical energy annual consumption in the 2014 sample by first and second homes recorded in EDP-SU survey.

Parameter description	Value
Sample	2014
No. all usable observations	2519
Median kWh, all observations	2153
No. observations, 1st or main home	2229
Median kWh, 1st or main home	2426
No. observations, 2nd home	314
Median kWh, 2nd home	473

A starting point for the determination of the low-occupancy consumption is the median consumption for reported second homes, equivalent to 1.3 kWh per day. This corresponds to the equivalent of 10 weeks of consumption per year of the average annual median consumption of designated first home occupants, which is also 1.3 kWh per day. By concentrating on these median values as the heuristic, we avoid excluding from the sample those designated second homes which have relatively large consumption levels and we avoid including in the sample those designated first homes which have extremely low consumption levels. This heuristic designates low-occupancy consumption as less than 10 weeks of the equivalent of the annual median consumption of all observations. There were two main factors in this choice: first, it results in estimates of the ratio of first and second home consumption that are approximately in line with estimates in the other studies cited, particularly that by Boogen [7]; second, it reflects the standard length of school and college vacation periods in Europe, and we believe is a primary factor in the adoption of second home consumption patterns. It can also be applied directly to the 2008 sample where the occupancy question was not included. This heuristic was also applied to the pseudo-panel data construction rule. The effect was to set a minimal annual consumption level for including an observation in 2008 at 590 kWh and in 2014 at 414 kWh. The 2014 figure is lower because consumption in all percentiles of the distribution fell between 2008 and 2014. These minimal annual consumption levels are similar to but slightly more conservative than the outlier exclusion values determined in Boogen's study.

Now, consider the upper tail of the consumption distribution. As with many consumer goods and services, the distribution of individual household consumption is heavily skewed in the positive direction with most households clustered around a moderate level of consumption (the 2008 median annual consumption over all households is 3072 kWh) and a long tail of higher consumption levels. However, there is a very small number of households, i.e., six households in 2008 and six households in 2014, with extremely high consumptions up to 20–40 times the median. In addition, and for reasons which are not clear, these few extremely high consumption levels are concentrated in one region, Litoral, which accounts for two-thirds of them. Including too many unlikely high observations biases the efficiency score downwards. Therefore, we excluded the electrical energy consumption of households above the 99.75 percentile of observations in 2008, i.e., those above 70,475 kWh per year, and in 2014, i.e., those above 36,244 kWh per year. This consideration also led to the exclusion of cohort 71 in the pseudo-panel data for the Litoral region with subscribed

demand band 6 and a family size less than three, where these extremely high observations are concentrated. The summary tables for all samples are shown below, in Tables 10–13.

Table 10. The 2008 sample’s summary statistics for individual households after data filtering (non-binary variables only).

Variable	Units	Obs	Mean	Std. Dev.	Min	Max
Subscribed demand	kVA	2314	10.37	7.64	1.15	41.40
Consumption	kWh	2314	5440.34	6421.49	592.00	65,778.00
Family size	Number	2314	3.13	1.26	1.00	12.00
Net income 2014 prices	Euros per month in 2014 prices	2314	1687.87	1016.04	196.01	3230.97
Demand charge 2014 prices	Euros per month in 2014 prices	2314	16.48	9.72	6.36	33.71
Energy charge 2014 prices	Euros per kWh in 2014 prices	2314	0.13	0.00	0.13	0.13
Percentage ownership of appliances	Percent	2314	0.65	0.14	0.13	1.00
Number of appliances owned	Number	2314	10.66	4.43	1.00	34.00

Table 11. The 2014 sample’s summary statistics for individual households after data filtering (non-binary variables only).

Variable	Units	Obs	Mean	Std. Dev.	Min	Max
Subscribed demand	kVA	2137	10.28	7.57	1.15	41.40
Consumption	kWh	2137	4103.63	4503.18	416.00	35,781.00
Family size	Number	2137	2.85	1.34	1.00	18.00
Net income 2014 prices	Euros per month in 2014 prices	2137	1431.30	912.20	242.50	3152.00
Demand charge 2014 prices	Euros per month in 2014 prices	2137	15.06	9.06	5.71	31.29
Energy charge 2014 prices	Euros per kWh in 2014 prices	2137	0.15	0.00	0.15	0.15
Percentage ownership of appliances	Percent	2137	0.64	0.16	0.06	1.00
Number of appliances owned	Number	2137	8.74	4.05	1.00	34.00

Table 12. The 2008 sample’s summary statistics for cohort means after data filtering (non-binary variables only).

Variable	Units	Obs	Mean	Std. Dev.	Min	Max
Subscribed demand	kVA	72	14.43	9.72	3.29	37.95
Consumption	kWh	72	8212.80	8468.67	1605.23	49,198.43
Family size	Number	72	2.84	1.05	1.00	4.33
Net income 2014 prices	Euros per month in 2014 prices	72	1737.06	421.17	784.03	2497.52
Demand charge 2014 prices	Euros per month in 2014 prices	72	20.71	10.14	6.36	33.71
Energy charge 2014 prices	Euros per kWh in 2014 prices	72	0.13	0.00	0.13	0.13
Percentage ownership of appliances	Percent	72	0.64	0.06	0.49	0.85
Number of appliances owned	Number	72	10.73	1.94	6.63	15.05

Table 13. The 2014 sample’s summary statistics for cohort means after data filtering (non-binary variables only).

Variable	Units	Obs	Mean	Std. Dev.	Min	Max
Subscribed demand	kVA	72	14.53	10.03	3.30	35.88
Consumption	kWh	72	6244.68	6054.07	1409.32	29,202.00
Family size	Number	72	2.84	1.08	1.55	4.59
Net income 2014 prices	Euros per month in 2014 prices	72	1570.80	485.42	727.23	2809.65
Demand charge 2014 prices	Euros per month in 2014 prices	72	19.00	9.51	5.71	31.29
Energy charge 2014 prices	Euros per kWh in 2014 prices	72	0.15	0.00	0.15	0.15
Percentage ownership of appliances	Percent	72	0.66	0.09	0.46	1.00
Number of appliances owned	Number	72	8.98	2.01	5.00	14.00

5. Estimation Results of the Models

We begin by summarizing the specifications that we used for the estimation of the theoretical models based on the data samples that we described in the previous section. Table 2 displays the calculation of elasticities from the equation specifications.

Equations for cross section samples:

Electrical energy consumption (kWh), e :

$$\begin{aligned} \ln e = & \alpha_0 + \alpha_1 \ln \text{net household income after demand charges} \\ & + \alpha_2 \ln \text{family size} + \alpha_3 \text{central heating ownership} \\ & + \alpha_4 \text{water heating ownership} \\ & + \alpha_5 \text{air conditioning ownership} \\ & + \alpha_6 \text{clothes drying ownership} \\ & + \alpha_7 \text{washing \& drying ownership} + v_e + u_e \end{aligned} \quad (40)$$

Subscribed power demand (kVA), q :

$$\begin{aligned} \ln q = & \beta_0 + \beta_1 \ln \text{net household income after demand charges} \\ & + \beta_2 \ln \text{family size} + \beta_3 \text{central heating ownership} \\ & + \beta_4 \text{water heating ownership} \\ & + \beta_5 \text{air conditioning ownership} \\ & + \beta_6 \text{clothes drying ownership} \\ & + \beta_7 \text{washing \& drying ownership} \\ & + \beta_8 \text{ownership percentage of appliance range} \\ & + \beta_9 \text{number of appliances owned} + v_q + u_q \end{aligned} \quad (41)$$

Equations for the pseudo-panel samples where cohorts are defined for the region, subscribed demand class and family size:

Electrical energy consumption (kWh), e :

$$\begin{aligned} \ln e = & \alpha_0 + \alpha_1 \ln \text{net household income after demand charges} \\ & + \alpha_2 \ln \text{unit price of energy} \\ & + \alpha_3 \text{central heating ownership} \\ & + \alpha_4 \text{water heating ownership} \\ & + \alpha_5 \text{air conditioning ownership} \\ & + \alpha_6 \text{clothes drying ownership} \\ & + \alpha_7 \text{washing \& drying ownership} + v_e + u_e \end{aligned} \quad (42)$$

Subscribed power demand (kVA), q :

$$\begin{aligned} \ln q = & \beta_0 + \beta_1 \ln \text{net household income after demand charges} \\ & + \beta_2 \text{central heating ownership} \\ & + \beta_3 \text{water heating ownership} \\ & + \beta_4 \text{air conditioning ownership} \\ & + \beta_5 \text{clothes drying ownership} \\ & + \beta_6 \text{washing \& drying ownership} \\ & + \beta_7 \text{ownership percentage of appliance range} \\ & + \beta_8 \text{number of appliances owned} + v_q + u_q \end{aligned} \quad (43)$$

The full set of regression results are available from the corresponding author. In the main text, we have selected the results from one of the models, Tables 14–16, i.e., the GLS-SUR generalized least squares estimation, although the results are very close over all three model types. We show the elasticities estimated at the sample mean and the mean efficiency scores for each of the 2008 and 2014 samples and the pooled pseudo-panel sample. We describe the elasticity estimates for all the models with results for both electrical energy consumption in Tables 17–19 and for subscribed power demand in Tables 20–22.

Table 14. Estimated parameters for the 2008 cross section sample.

GLS-SUR 2008	Equation	Equation
Variable	Energy Consumption kWh	Subscribed Power Demand kVA
Net income after demand charges	0.076 ***	0.151 ***
Family size	0.362 ***	0.094 ***
Central heating ownership	0.380 ***	0.342 ***
Water heating ownership	0.262 ***	0.155 ***
Air conditioning ownership	0.260 ***	0.160 ***
Clothes dryer ownership	0.308 ***	0.161 ***
Washer and dryer ownership	0.214 ***	0.190 ***
Ownership percentage of appliance range		0.108
Number of appliances owned		0.007 *
Constant	7.003 ***	0.595 ***
σ_u	0.752 ***	0.705 ***
σ_v	0.673 ***	0.521 ***
λ	1.117 ***	1.353 ***
N	2314	2314
R ²	0.1435	0.1374
chi ²	387.59 ***	346.45 ***
Mean efficiency	0.628	0.626
Breusch–Pagan test of cross equation error correlation	chi ² (1) = 784.116, Pr = 0.0000	

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.**Table 15.** Estimated parameters for the 2014 cross section sample.

GLS-SUR 2014	Equation	Equation
Variable	Energy Consumption kWh	Subscribed Power Demand kVA
Net income after demand charges	0.138 ***	0.226 ***
Family size	0.412 ***	0.149 ***
Central heating ownership	0.571 ***	0.416 ***
Water heating ownership	0.313 ***	0.219 ***
Air conditioning ownership	0.266 ***	0.140 ***
Clothes dryer ownership	0.306 ***	0.146 ***
Washer and dryer ownership	0.159	0.120
Ownership percentage of appliance range		0.021
Number of appliances owned		0.017 ***
Constant	6.293 ***	0.046
σ_u	0.448 **	0.609 ***
σ_v	0.766 ***	0.499 ***
λ	0.585 **	1.219 **
N	2137	2137
R ²	0.213	0.259
chi ²	579.48 ***	699.73 ***
Mean efficiency	0.742	0.663
Breusch–Pagan test of cross equation error correlation	chi ² (1) = 5 33.188, Pr = 0.0000	

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.**Table 16.** Estimated parameters for pooled pseudo-panel cohorts.

GLS-SUR Pooled	Equation	Equation
Variable	Energy Consumption kWh	Subscribed Power Demand kVA
Net income after demand charges	0.238	0.873 ***
Energy price	−0.889 ***	
Central heating ownership	2.869 ***	2.783 ***
Water heating ownership	0.563 *	0.695 *

Table 16. Cont.

GLS-SUR Pooled	Equation	Equation
Variable	Energy Consumption kWh	Subscribed Power Demand kVA
Air conditioning ownership	0.510 ***	0.559 ***
Clothes dryer ownership	1.591 ***	1.264 ***
Washing and dryer ownership	2.907 **	2.036 **
Ownership percentage of appliance range		−1.888 **
Number of appliances owned		0.006
Constant	3.817329	−4.051 ***
σ_u	0.698 ***	0.685 ***
σ_v	0.229 ***	0.288 ***
λ	3.038 ***	2.376 ***
N	142	142
R ²	0.567	0.529
chi ²	190.14	173.52
Mean efficiency	0.629	0.620
Breusch–Pagan test of cross equation error correlation	chi ² (1) = 104.209, Pr = 0.0000	

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 17. The 2008 cross section's electrical energy consumption.

Sample	Elasticity with Respect to	
2008 cross section of individual households	Net income before demand charges	0.0771
	Elasticity with respect to demand charges, euros/month	−0.0011
	Elasticity with respect to energy price, euros/kWh	zero variation
	Mean behavioral efficiency	0.628

Table 18. The 2014 cross section's electrical energy consumption.

Sample	Elasticity with Respect to	
2014 cross section of individual households	Net income before demand charges	0.1397
	Elasticity with respect to demand charges, euros/month	−0.0017
	Elasticity with respect to energy price, euros/kWh	zero variation
	Mean behavioral efficiency	0.742

Table 19. Pooled cohort sample's electrical energy consumption.

Sample	Elasticity with Respect to	
Pooled cohort sample	Elasticity with respect to net income before demand charges	0.2406
	Elasticity with respect to demand charges, euros/month	−0.0026
	Elasticity with respect to energy price euros/kWh	−0.889
	Mean behavioral efficiency	0.629

Table 20. The 2008 sample's subscribed power demand.

Sample	Elasticity with Respect to	
2008 cross section of individual households	Net income before demand charges	0.1531
	Elasticity with respect to demand charges, euros/month	−0.0021
	Mean behavioral efficiency	0.626

Table 21. The 2014 sample's subscribed power demand.

Sample	Elasticity with Respect to	
2014 cross section of individual households	Net income before demand charges	0.2288
	Elasticity with respect to demand charges, euros/month	−0.0028
	Mean behavioral efficiency	0.663

Table 22. Pooled sample's subscribed power demand.

Sample	Elasticity with Respect to	
Pooled cohort sample	Elasticity with respect to net income before demand charges	0.8824
	Elasticity with respect to demand charges, euros/month	−0.0094
	Mean behavioral efficiency	0.62

5.1. Regression Results

We begin with comments on the full set of regression results. Each model consists of two equations, one for electrical energy consumption (kWh) and one for subscribed power demand (kVA).

Overall, the results, in terms of coefficient magnitude, statistical significance, and error decomposition, are consistent across all estimation methods. In particular, the 2SLS results are very close to the MLE-SFA and GLS-SUR results. This should not be surprising since the instrumental variables used to account for the endogeneity of the net income after demand charges are dominated by the net income before demand charges and these two variables are highly correlated. In general, as well, the mean efficiency estimated in each of the three models is highly consistent in 2008, in 2014, and again in the pseudo-panel. However, it is important to observe that the estimated parameters and efficiency estimates, although consistent across estimation models, differ importantly across the samples, with a marked difference between the 2008 and 2014 samples.

Most variables are consistently significant at the five percent and one percent significance levels, with only the variables for the ownership percentage of appliances ranging and the number of appliances owned sometimes being insignificant in the subscribed power demand equation, (they did not feature in the electrical energy consumption equation). Each of the models produces very similar results. To keep the analysis concise, we need select only one set of results for each market and time period as representative. We report the SUR-GLS results for three reasons. First, they are superior to the MLE-SFA in the fact that they permit the two demand equations to be interrelated, whereas the MLE-SFA results treat them as unrelated equations. Second, they are superior to the IV-2SLS results for the same reason, while the endogeneity allowed for in the IV-2SLS results does not produce significantly different estimates. Thirdly, the SUR-GLS estimators are consistent, which is a required property for the application of the concentrated likelihood maximization in the application of the Fan et al. [29] procedure.

Goodness of fit statistics are reported for the GLS-SUR estimates and are highly significant. In all cases, the inefficiency component of the composed error (measured by $\lambda = \sigma_u / \sigma_v$) is important and significant. In the GLS-SUR estimates, the Breusch–Pagan test confirms the significance of the cross-equation error correlation. We could therefore consider the GLS-SUR results to be the most robust estimates, but the other estimation methods give very similar results.

We turn now to the individual explanatory variables. They all have the expected signs. The net income after demand charges is consistently estimated and statistically significant for both electrical energy consumption and subscribed power demand in both of the cross section samples, but is not so important in the pseudo-panel sample. The demand charge impact is included in this variable but its elasticity can be separated out. Family size is highly significant in the cross section samples across all methods (it does not appear in the pseudo-panel since it is used as one of the cohort-defining characteristics)

but is quantitatively more important for electricity consumption than for subscribed power demand. The energy charge can only appear in the pseudo-panel sample for the electricity consumption since it does not vary across cross sections, and its elasticity is negative in each case and strongly significant in the GLS-SUR and 2SLS estimates. The individual appliance ownership variables that are identified with high power ratings are strongly significant across all estimation methods and across all samples.

We can now turn to the key elasticity and mean efficiency results shown in Tables 17–22. These confirm that the elasticities are consistently estimated and we note that the parameter estimates embedded in the elasticities are all significantly different from zero at the five percent significance level so that we can have considerable confidence in the elasticity estimates. For the income effects, we separate out the income elasticity with respect to the net income before demand charges and the price elasticity with respect to the demand charges alone to obtain two views of the role of the income effect.

In the case of electrical energy consumption (e kWh), seen in Tables 17–19, the income elasticity for 2008 is between 0.069 and 0.087 depending on the estimation method used, and is 0.077 for the GLS-SUR model. This translates to a price elasticity with respect to demand charges (euros/month) of -0.001 across all estimation methods. There is therefore a considerable scope for raising demand charges before there would be a discernible effect on residential electricity consumption. In comparison, the corresponding values for 2014 have almost doubled in size. The income elasticity for 2014 is between 0.136 and 0.147 depending on the estimation method used, and is 0.140 for the GLS-SUR model, while the price elasticity with respect to demand charges is -0.002 across all estimation methods. There are three clear implications of these results. First, the macroeconomic changes in Portugal following the global financial crisis of 2008 have had considerable effects on microeconomic behavior as shown in our case by the doubling in the sensitivity of electricity consumption to income changes. Secondly, there is considerable scope for raising demand charges with only a very minor impact on electricity consumption. Thirdly, and possibly most important from the point of view of energy conservation and emissions reduction, it follows that it would require massive rises in demand charges to have a large effect in reducing residential energy consumption. Therefore, the consequential impact on consumers' incomes of market-based environmental policies using the required large increases in demand charges could be an important limiting factor on the scope of these policies. Put more simply, the adverse income distribution effects of using much higher demand charges to induce energy conservation will be large.

It is possible to measure an additional price elasticity with respect to the energy charge (euros/kWh) in the pooled pseudo-panel sample where there is a variation over time and therefore across cohort mean observations in the energy charge. For residential electricity consumption, the price elasticity with respect to the energy charge is between the values of -0.672 and -1.076 depending on the estimation method used, and is -0.889 for the GLS-SUR model. This is much higher than the demand charge elasticity and does suggest that there is potential for a market-based energy conservation and emissions reduction policy. Nevertheless, there is an important caveat to be noted. The pooled pseudo-panel of cohorts covers two widely spaced periods: 2008 and 2014. However, there are several major changes that occurred between 2008 and 2014: certainly, the energy charges change, but there is also the fact that the weather may have been different; there was the passage of time with its potential impact on the technological progress in appliance design; the electricity market was opened to competition; and VAT was increased. The change in energy charges is measurable only as a discrete difference between the level of the energy charge in euros/kWh in 2008 and its value in 2014. It is a single discrete shift, because there is zero or negligible variation in energy charges for individual cohorts of consumers. But the passage of time is a single discrete shift as well, observable as the difference in mean residential consumption in the two years. The microeconomic policy intervention of increased retail competition is also a discrete shift, as is the macroeconomic policy intervention of the increase in VAT.

We examined weather differences in the two years without obtaining meaningful results. Cooling degree days' (cdd) data are at a lower level in 2008 than 2014, requiring lower use of air conditioning, but heating degree days' data (hdd) are higher in 2008 than 2014, requiring greater use of central heating and water heating. Separately entered into the pooled sample, the effects of the two measures offset each other. Consequently, the weather variation between 2008 and 2014 is too indefinite to measure its impact. In any case, from the answers to the questionnaires, we found that the rate of the use of heating systems was lower in 2014.

Therefore, in the pooled sample, we are left with the other four sources of discrete shifts between 2008 and 2014, which means that we can measure the price elasticity of residential electricity consumption with respect to time, increased competition, VAT increase and energy charges combined. This is the result that we show as the 'energy charge elasticity' in the pooled sample. It is therefore likely to be an overestimate of the underlying behavioral price elasticity but it is impossible to say how much of an overestimate.

Finally, consider the behavioral efficiency of residential electricity consumption. Depending on the estimation method used, the mean behavioral efficiency for 2008 is between 62.6 percent and 62.9 percent, and is 62.8 percent for the GLS-SUR model. These are very consistent results even though there is considerable difference in the estimation approaches. There is similar consistency across estimation methods in 2014 with the mean behavioral efficiency lying between 74.0 percent and 74.6 percent, and it is 74.2 percent for the GLS-SUR model. The noticeable features are that there has been a large improvement in behavioral efficiency between 2008 and 2014, possibly related to the strong conservation policy initiatives put in place by the Portuguese government, the energy regulator, ERSE and the major retailer EDP, but also possibly due to the significant loss of family income between 2008 and 2014. There remains, however, considerable scope for improvements in the behavioral efficiency of residential electricity consumption since these results indicate a level of behavioral efficiency substantially lower than those reported in our survey of other studies in Table 1.

We now turn to the subscribed power demand results, Tables 20–22, an aspect which does not seem to have been the subject of other research in this field. Three parametric results are reported in each table: the income elasticity of subscribed power demand with respect to net income before demand charges, the price elasticity with respect to demand charges, and the mean behavioral efficiency. The income elasticity of subscribed power demand lies between 0.120 and 0.153 in 2008 (0.153 for the GLS-SUR model), and between 0.177 and 0.229 in 2014 (0.229 for the GLS-SUR model), so that it displays the same tendency over all models to increase during the macroeconomic upheaval in Portugal after the global financial crisis, as we found in the case of electricity consumption. The pooled cohort sample suggests higher values for the income elasticity, between 0.377 and 0.882 (0.882 for the GLS-SUR model). We can see the reason for this in the summary statistics for the final samples in Tables 12 and 13. Between 2008 and 2014, in line with the fall in real net family income, the ownership of appliances also declined among households at the top end of the appliance ownership range. Both the maximum number of appliances owned and the maximum percentage of ownership of appliances declined in line with the drop in family incomes consequential to the macroeconomic shock to the economy of Portugal between 2008 and 2014. This enhanced impact on subscribed power demand from the income shock is evidence again of the feedback on the microeconomic behavior of the macroeconomic direction of the economy.

The price elasticity of subscribed power demand with respect to demand charges is very low, varying between -0.002 and -0.003 across all the different samples and estimation methods. Behavioral efficiency in subscribed demand leaves room for large improvements in efficient behavior, ranging between 62 and 66 percent over all samples and estimation methods. However, the low-price elasticity of subscribed power demand with respect to demand charges indicates that very large increases in demand charges

would be needed to eliminate this inefficiency if a market-based incentive policy is adopted, and these must have consequent income distribution effects.

Finally, we now consider appliance response results. In Table 2, we showed the formula for the percentage change effect as 100 times the proportionate or relative effect on the dependent variable when a dummy variable changes from a value of zero to one. However, this is a biased estimator unless the regression coefficients are known exactly. Kennedy [34] suggests instead the following Approximate Unbiased Percentage Effect Estimator:

$$pe_{e, \text{appliance ownership}_k} = 100 \times [\exp(\hat{\alpha}_k - (1/2)V(\hat{\alpha}_k)) - 1] \quad (44)$$

Here, $V(\hat{\alpha}_k)$ is the estimated variance in the regression coefficient on the dummy variable. van Garderen and Shah [35] show that the bias in Kennedy's estimator is negligible. There is a similar adjustment for power demand:

$$pe_{q, \text{appliance ownership}_k} = 100 \times [\exp(\hat{\beta}_k - (1/2)V(\hat{\beta}_k)) - 1] \quad (45)$$

These percentage change effects are shown in Table 23. Alongside each percentage effect is its standard error calculated using van Garderen and Shah's Approximate Unbiased Variance of Percentage Effect Estimator.

Table 23. Percentage effects of changes in high-power rating appliance ownership statuses, based on Kennedy [34] and the corresponding standard errors (SEs), based on van Garderen and Shah [35].

Appliance Type	Percentage of Change in the Dependent Variable from the Change in the Ownership Binary Variable from 0 to 1			
	Electricity Consumption		Power Demand	
2008	percent change effect	SE	percent change effect	SE
Central heating ownership	45.69	12.52	40.40	10.04
Water heating ownership	29.84	5.85	16.62	4.50
Air conditioning ownership	29.64	5.08	17.29	3.99
Clothes dryer ownership	35.91	5.00	17.43	3.83
Washer and dryer ownership	23.47	9.28	20.75	7.53
2014	percent change effect	SE	percent change effect	SE
Central heating ownership	76.23	14.61	51.29	9.70
Water heating ownership	36.61	5.87	24.41	4.26
Air conditioning ownership	30.37	5.54	14.94	3.90
Clothes dryer ownership	35.65	5.74	15.62	3.98
Washer and dryer ownership	16.62	12.41	12.37	9.18

We see immediately that when a typical individual household acquires an appliance with a high-power rating for the first time, its electricity consumption and its power demand are likely to increase substantially. For example, the installation of electric central heating in 2008 corresponded to a 45 percent higher level of consumption in those households with electric central heating compared to those without, and a 40 percent increase in their power demand. The 2014 figures are even higher (although from a lower base level of mean consumption than that in 2008), a 76 percent higher electricity consumption and 51 percent higher subscribed power demand from the ownership of electric central heating. As Table 23 shows, these percentage effects are several times greater than their standard errors (SEs) for both electricity consumption and power demand for all of the high-power rating appliances in the 2008 sample and similarly in 2014, except for washing and drying combined. At present, many consumers in Portugal do not use central heating systems, electric water heating or air conditioning. An increase in the rate of ownership of these types of appliances would imply a significant rise in electricity consumption. This could be an interesting opportunity for the sector to improve the utilization of the current electrical energy network capacity, following the recent reductions in consumption. At the same time, the necessary investments in equipment and consumption could be costly for consumers.

5.2. Individual Welfare Effects and Behavioral Agents

We can use these empirical results to compare the welfare effects of price changes for rational and behavioral agents. Behavioral inefficiency is prevalent across the samples and Pigouvian prices such as carbon taxes are often advocated to correct inefficient energy consumption, according to Timilsina [36]. Following Farhi and Gabaix [19], we saw how to calculate the welfare effects of price changes on rational and behavioral agents using the conventional and amended Roy identities, Equations (29), (30), (33) and (34). In this analysis, we contrasted the agent's perceived and experienced utilities, which are identical for rational agents but different for behavioral agents. The objective of a policy-based price increase for energy or power demand is to move the behavioral agent closer to the optimum of the rational agent, i.e., to incentivize energy consumption and power demand to move closer to the efficient frontier. The first key component in the calculation is the behavioral wedge calculated by the proportion of the commodity prices which are internalized by the agent. We measured the electrical energy and power demand behavioral wedges as $(1 - BE_e)\tilde{p}_e$ and $(1 - BE_q)\tilde{p}_q$. The empirical results, including price and income elasticities, allow us to calculate Equations (33) and (34) to produce the results shown in Table 24. Because of the limited data variation in the energy price across the cross section samples, we are only able to measure the welfare effect of an energy price change for the pooled sample. On the other hand, because the power demand categories are used to define the cohorts in the pooled sample, we are only able to measure the welfare effects of a power demand price change for the two cross section samples. However, the results are interesting for policy reasons.

Table 24. Welfare effects for rational and behavioral agents of a doubling of prices.

Comparative Static Effect		Sample Results		
Energy price change: Equation (33) Roy identity		2008	2014	Pooled sample
$\left(\frac{\partial v/\partial \tilde{p}_e}{\partial v/\partial (\tilde{y})}\right)d\tilde{p}_e$	Rational agent, euros/year	N/A	N/A	−1156.6
$\left(\frac{\partial v/\partial \tilde{p}_e}{\partial v/\partial (\tilde{y})}\right)d\tilde{p}_e$	Behavioral agent, euros/year	N/A	N/A	−781.02
Welfare offset for behavioral agent, euros/year		N/A	N/A	375.58
Demand charge change: Equation (34) Roy identity		2008	2014	Pooled sample
$\left(\frac{\partial v/\partial \tilde{p}_q}{\partial v/\partial (\tilde{y})}\right)d\tilde{p}_q$	Rational agent, euros/year	−2050.77	−1857.80	NA
$\left(\frac{\partial v/\partial \tilde{p}_q}{\partial v/\partial (\tilde{y})}\right)d\tilde{p}_q$	Behavioral agent, euros/year	−2050.25	−1856.41	NA
Welfare offset for behavioral agent, euros/year		0.52	1.39	NA

We can now relate the results in Table 24 to the diagrammatic effects illustrated earlier in the paper in Figure 1 in Section 3. A doubling of the energy price change leads to a welfare loss for the rational agent of €1156.6 per year, but for the behavioral agent the welfare loss is the lower value of €781.02. This is because the welfare loss offset, illustrated in Figure 1 in Section 3 of the paper by the money metric distance between the experienced utility indifference curves through B and through C, is measured at €375.58 per year. This represents the welfare gain to the behavioral agent in experienced (rather than perceived) utility by the incentive to be located closer to the efficient frontier of energy consumption.

The second key component of the welfare effect for behavioral agents based on Roy identities consists of the Slutsky substitution effects. These are large for the impact of an energy price change in the pooled sample, but extremely small for the impact of a change in power demand charges in cross section samples. In the bottom half of Table 24, we can see that the effect of a doubling in the power demand charge of Δp_q has roughly the same negative effect on the welfare of both rational and behavioral agents; the offset to the welfare loss incentivizes the behavioral agent to move closer to the frontier amounts

to only 0.52 to 1.39 euros per year. However, in the upper half of Table 24, we see a much stronger effect from the imposition of a doubling in the price of electrical energy, Δp_e . Using the pooled sample results, the welfare effects on the rational and behavioral agents differ by 375.58 euros per year, so that this represents the reduction in the loss of individual welfare from the increased p_e that would be experienced by behavioral agents moving closer to the efficient frontier. We conclude from this that changes in the price of electrical energy will have much more important incentive effects on behavioral consumers than changes in the charges for power demand. For policy makers seeking to incentivize reduced energy consumption, this offers a way to nudge energy consumption close to the efficient frontier. It is clear, however, that even for behavioral agents, and even more for rational agents, the Roy identity welfare losses from increments to energy prices could amount to a substantial share in income per person. This highlights a critical policy issue: the trade-off between equity and efficiency. Economic models emphasize the efficiency gains from reduced energy emissions that are available through pricing policies, but there may be acute political difficulties in implementing these if they involve large welfare losses. This point is made by Farhi and Gabaix, and also argued by Romero-Jordan and del Rio (2022) [9].

6. Conclusions

Understanding the behavior of electricity-consuming households and their efficiency in energy decision-making is essential for sustainability

The dataset consisted of two large-scale household surveys carried out in Portugal through stratified interview processes. Usable data after excluding incomplete responses and censoring the consumption data to exclude low-occupancy households amounted to 2314 observations in 2008 and 2137 observations in 2014. We also combined these into a pseudo-panel sample of 141 cohorts defined by the subscribed power demand band, family size and region. We measured elasticity responses for net income, demand charges, energy charges and the impact of appliance ownership status and estimates of behavioral efficiency.

The key result is that there was significant inefficiency in electrical energy consumption and in subscribed power demand. Therefore, there is considerable room for behavioral efficiency improvements over the relatively low levels achieved in order to enhance sustainability in residential energy consumption. We demonstrated how this could be explained by the responses of behavioral rather than rational agents, and we measured the behavioral wedge in electricity market behavior. Improved efficiency should be feasible given the observed rise in behavioral efficiency in electricity consumption from around 63 percent in 2008 to 74 percent in 2014. The Portuguese authorities and the EDP (Energias de Portugal) have carried out policies aimed at educating consumers about their usage, i.e., ‘nudges’. There is room for applying this further in electricity consumption and in the investment decisions about appliances that contribute to subscribed power demand. In particular, we noted the very high impact associated with the decision to acquire high-power rating appliances such as air-conditioning and space heating. In this context, smart grid technologies and real-time pricing will become even more important. We direct the reader to the study by Decker [18] for a generalization of this to other industries that have public utility characteristics.

There is evidence from the pooled cohort sample of relatively high energy price elasticity which could be potentially useful for policy purposes, but we saw that this estimate is difficult to separate from the major changes which impacted the consumer from the macroeconomic shocks to the economy. An exception to the negative shock effect is the liberalization of the market to enhance competition, which may have contributed to the relatively high energy price elasticity found in the pooled sample. However, an important element of the consumers’ energy bills is the level of the demand charge. The price elasticity of both electricity consumption and subscribed power demand with respect to demand charges is very low, between -0.001 and -0.002 . While this gives a scope to electricity retail companies for large increases in the demand charges without affecting demand strongly,

it does indicate that very large increases in demand charges would be required to have an effect on electricity consumption and power demand with almost certainly serious consequences for the distribution of income amongst households, particularly at the lower end of the scale. In addition, while the switch made by households to electricity-using appliances for central heating and water heating as well as air conditioning is a necessary component of emissions policy and sustainability, it implies very high percentage increases in electricity consumption and power demand. For environmental reasons, policy makers may want to encourage the trade-off between fossil fuel use and electrification.

We successfully estimated the indirect welfare effects of price changes for electrical energy and power demand on non-rational or behavioral agent consumers. These indicate that emission taxes on electricity prices could have a positive offset to the consequent loss of individual welfare where the positive offset results from moving non-rational or behavioral agents closer to the efficient demand frontier. However, the size of the price increases would be likely to have possibly severe negative effects on the distribution of consumer incomes across the population. This could mean that the scope for using pricing policy alone to drive sustainability may be limited by issues of fairness and political acceptability. Further research is needed to understand how these findings relate to the broader economic and policy context, especially considering the potential for revenue recycling or redistribution to address equity concerns. Additionally, further research is needed to understand how these effects would appear in other countries and populations.

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Appendix A

The derivation of these results follows Sydsaeter et al. [37]. The general problem in Equation (6) in the text, replacing the equation constraints with inequality constraints, is:

$$U^* = v(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') = \left\{ \max_{\mathbf{x}, e, q \geq 0} U(\mathbf{x}') : \mathbf{p}'\mathbf{x} \leq \tilde{y} - \tilde{p}_e e - \tilde{p}_q q; e \geq e^{\min}(\mathbf{z}_e'); q \geq q^{\min}(\mathbf{z}_q') \right\} \quad (\text{A1})$$

The Lagrangean function for the problem in Equation (A1) is

$$\mathcal{L} = U(\mathbf{x}') + \theta \left((\tilde{y} - \tilde{p}_e e - \tilde{p}_q q) - \mathbf{p}'\mathbf{x} \right) + \mu \left(e - e^{\min}(\mathbf{z}_e') \right) + \zeta \left(q - q^{\min}(\mathbf{z}_q') \right) \quad (\text{A2})$$

The first-order necessary conditions are:

$$\partial \mathcal{L} / \partial x_j = \partial U / \partial x_j - \theta p_j \leq 0; j = 1 \dots m; x_j \partial \mathcal{L} / \partial x_j = 0 \quad (\text{A3})$$

$$\partial \mathcal{L} / \partial e = -\theta \tilde{p}_e + \mu \leq 0; e \partial \mathcal{L} / \partial e = 0 \quad (\text{A4})$$

$$\partial \mathcal{L} / \partial q = -\theta \tilde{p}_q + \xi \leq 0; q \partial \mathcal{L} / \partial q = 0 \quad (\text{A5})$$

The Envelope theorem states that the effect on the maximal indirect utility given by the value function $U^* = v(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q')$ of a change in an exogenous parameter such as one of the prices or one of the constraint constants is the derivative of the Lagrangean function with respect to that parameter, holding all the endogenous variables at their optimal values.

Therefore, the Envelope theorem states that the multipliers are interpreted as the marginal utility of nominal income, the marginal disutility of the constraint on minimum energy input, and the marginal disutility of the constraint on minimum subscribed power demand:

$$\theta = \partial v / \partial (\tilde{y}) ; \mu = -\partial v / \partial e^{\min}; \xi = -\partial v / \partial q^{\min} \quad (\text{A6})$$

The solution produces the demand functions for each of the commodities along with energy usage and subscribed power demand:

$$x_j = x_j(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') ; j = 1, \dots, m \quad (\text{A7})$$

$$e = e(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') \quad (\text{A8})$$

$$q = q(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') \quad (\text{A9})$$

If the Lagrangean function is concave in $(\mathbf{x}, e, q)$, then these conditions are also sufficient.

At an interior optimum when all the constraints are satisfied, we can write these first-order conditions as follows, where each of the marginal utility terms is measured relative to the marginal utility of income, so that each contribution to indirect utility is expressed as a money metric, i.e., in money terms. In addition, we rewrite the marginal disutility of electrical energy input and the marginal disutility of power demand as positive terms representing their respective contributions to indirect utility by supporting the consumption of utility-producing goods and services. In this way, we recognize that the additional purchase of energy or power demand can contribute to the consumption of more of the other commodities.

$$p_j = \frac{\partial U / \partial x_j}{\partial v / \partial (\tilde{y})} \quad (\text{A10})$$

$$\tilde{p}_e = \frac{\partial v / \partial e}{\partial v / \partial (\tilde{y})} \quad (\text{A11})$$

$$\tilde{p}_q = \frac{\partial v / \partial q}{\partial v / \partial (\tilde{y})} \quad (\text{A12})$$

From the solution, we can also use the Envelope theorem to evaluate the effects on the individual's welfare of changes in the exogenous prices. These are the expressions for Roy's identity and indicate the marginal effect on the consumer's maximal indirect utility of a change in each of the prices relative to the marginal utility of income. The interpretation in the study by Sydsaeter et al. [37] is that for a small price change, the loss of welfare or real income or utility is approximately equal to the change in price multiplied by the negative of the quantity demanded.

$$\left(\frac{\partial v / \partial p_j}{\partial v / \partial (\tilde{y})} \right) dp_j = -x_j(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') dp_j \quad (\text{A13})$$

$$\left(\frac{\partial v / \partial \tilde{p}_e}{\partial v / \partial (\tilde{y})} \right) d\tilde{p}_e = -e(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') d\tilde{p}_e \quad (\text{A14})$$

$$\left(\frac{\partial v / \partial \tilde{p}_q}{\partial v / \partial (\tilde{y})} \right) d\tilde{p}_q = -q(\mathbf{p}', \tilde{p}_e, \tilde{p}_q, \tilde{y}, \mathbf{z}_e', \mathbf{z}_q') d\tilde{p}_q \quad (\text{A15})$$

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